CV

CostVision

Manufacturing Should-Cost Intelligence Platform

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Flange with Central Boss (6-Hole Bolt Pattern)

Commodity: MACHINING · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 5 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥149.77	**4.5%**	**55.4%**	**21.4%**	**7.4%**

Model Confidence: **Low** · 5 traced operations · 12 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 63.9 cm³; mass for the selected material family 0.502 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: machining

Alloy / material: 1045 / C45 / 080M46 (Medium-Carbon Steel) · Net weight: 0.502 kg (measured)

Material utilisation: 50% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 5

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥6.72	4.5%	
2. Process (Machine)	¥83.02	55.4%	
3. Direct Labour	¥32.11	21.4%	
4. Tooling (amortised)	¥0.68	0.5%	
5. Packaging	¥0.63	0.4%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥0.81	0.5%	
Factory Cost	**¥123.98**	**82.8%**	
7. Overhead (SG&A)	¥14.70	9.8%	
Subtotal	**¥138.68**	**92.6%**	
8. Supplier Margin	¥11.09	7.4%	
TOTAL SHOULD-COST	**¥149.77**	**100.0%**	

§2 — Commercial Parameters

Overhead Rate	**12.0%**	Supplier Margin	**8.0%**
Packaging / part	**¥0.63**	Logistics / part	**¥0.81**
Total Tooling Cost	**¥135746.61**	Amortisation Volume	**200,000 parts**

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	**mat-steel1045**	ID	
Grade / Specification	**1045 / C45 / 080M46 (Medium-Carbon Steel)**		UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. | Regional adj. ×0.880 (millSteel)
Region	**China**		
Net Finished Weight	**0.5020 kg**	kg	Weight in finished part
Gross Weight (stock)	**1.0040 kg**	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	**0.5020 kg**	kg	Gross - Net
Material Utilisation	**50.0%**	%	Benchmark: machining 60–75 % (stock removal)
Material Price	**¥7.57**	CNY /kg	UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. | Regional adj. ×0.880 (millSteel)
Scrap Recovery Price	**¥1.75**	CNY /kg	
Gross Material Cost	**¥7.60**	CNY	Gross × price/kg
Scrap Credit	**-¥0.88**	CNY	Scrap × recovery price
NET RAW MATERIAL COST	**¥6.72**	CNY	Gross - scrap credit + consumables
Data Confidence	**Medium**		
Effective Date	**2026-07**		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Setup (amortised)	CNC Lathe (2-axis)	¥185.20	0.60	1	100.0 %	**¥1.85**
Turning — +X (8 faces)	CNC Lathe (2-axis)	¥185.20	10.21	1	85.0%	**¥37.06**
Turning — +Z (3 faces)	CNC Lathe (2-axis)	¥185.20	3.81	1	85.0%	**¥13.85**
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	CNC Drilling Centre	¥136.82	7.22	1	85.0%	**¥19.36**

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	CNC Lathe (2-axis)	¥185.20	3.00	1	85.0%	**¥10.89**
TOTAL						**¥83.02**

4B Labour Detail

Operation	Labour Grade	Rate / hr	Manning	Lab. min	Eff %	Labour Cost	Op Total
Setup (amortised)	Skilled Machinist	¥71.49	1	0.60	100.0%	¥0.71	**¥2.57**
Turning — +X (8 faces)	Skilled Machinist	¥71.49	1	10.21	92.0%	¥13.22	**¥50.28**
Turning — +Z (3 faces)	Skilled Machinist	¥71.49	1	3.81	92.0%	¥4.94	**¥18.79**
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	Skilled Machinist	¥71.49	1	7.22	92.0%	¥9.35	**¥28.71**
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	Skilled Machinist	¥71.49	1	3.00	92.0%	¥3.89	**¥14.78**
TOTAL						**¥32.11**	**¥115.13**

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Boss	Ø30.0 × 8	1	Turning (external Ø)	0.46	¥1.95	Yes
Boss	Ø80.0 × 12	1	Turning (external Ø)	0.54	¥2.29	Yes
Hole / bore	Ø6.0 × 12 blind	6	Drilling (blind)	1.68	¥7.12	Yes
Face	4857 mm²	1	Face milling (facing)	0.81	¥3.42	Yes
Face	4150 mm²	1	Face milling (facing)	0.72	¥3.05	Yes
Face	707 mm²	1	Face milling (facing)	0.29	¥1.22	Yes
TOTAL costed		**11**		**4.49**	**¥19.05**	**Yes**

§5 — Machine Rate Buildup

CNC Lathe (2-axis) [mach-lathe-cnc] · Confidence: Low · UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥199095.02	¥62.22	4,000 hr/yr available
Maintenance	¥109502.26	¥34.22	
Energy	¥55085.58	¥17.21	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥109502.26	¥34.22	
Finance Cost	¥79638.01	¥24.89	
TOTAL — CNC Lathe (2-axis)	¥592642.14	¥185.20	Effective hours: 3200/yr

CNC Drilling Centre [mach-drill] · Confidence: Low · UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥139366.52	¥43.55	4,000 hr/yr available
Maintenance	¥59728.51	¥18.67	
Energy	¥49577.02	¥15.49	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥79638.01	¥24.89	
Finance Cost	¥69683.26	¥21.78	
TOTAL — CNC Drilling Centre	¥437812.32	¥136.82	Effective hours: 3200/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	0.8360	£/kg	UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. | Regional adj. ×0.880 (millSteel)	mat-steel1045	Medium

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.scrapRecoveryPricePerKg	0.1936	£/kg	UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. | Regional adj. ×0.880 (millSteel)	mat-steel1045	Medium
Setup (amortised).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Setup (amortised).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Turning — +X (8 faces).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Turning — +X (8 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Turning — +Z (3 faces).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Turning — +Z (3 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].machineRatePerHr	15.1182	£/hr	UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-drill	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥149.77 ± 20%	¥121.54	¥147.96	¥181.45	wide	15.9%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 10 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
Import duty and international freight (regional table is Ex-Works).
Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Setup (amortised): Machine Rate	¥185.20/hr	-5.1%	+5.1%	**¥15.40**
Turning — +X (8 faces): Machine Rate	¥185.20/hr	-5.1%	+5.1%	**¥15.40**
Turning — +Z (3 faces): Machine Rate	¥185.20/hr	-5.1%	+5.1%	**¥15.40**
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]: Machine Rate	¥185.20/hr	-5.1%	+5.1%	**¥15.40**
Turning — +X (8 faces): Cycle Time	0.17hr	-4.1%	+4.1%	**¥12.16**
Setup (amortised): Labour Rate	¥71.49/hr	-2.6%	+2.6%	**¥7.77**
Turning — +X (8 faces): Labour Rate	¥71.49/hr	-2.6%	+2.6%	**¥7.77**
Turning — +Z (3 faces): Labour Rate	¥71.49/hr	-2.6%	+2.6%	**¥7.77**

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥7.63	¥150.95	¥115.59	¥1.23	¥19.60	**¥295.01**	¥0.56	**¥307.57**	+105%
Germany (EUR)	¥7.86	¥158.50	¥186.81	¥1.30	¥27.76	**¥382.22**	¥0.65	**¥398.18**	+166%
France (EUR)	¥7.78	¥138.88	¥134.27	¥1.14	¥21.08	**¥303.15**	¥0.65	**¥316.15**	+111%
Spain (EUR)	¥7.63	¥132.84	¥84.65	¥1.09	¥15.30	**¥241.50**	¥0.65	**¥252.17**	+68%
Poland (PLN)	¥7.40	¥108.69	¥52.54	¥0.89	¥10.26	**¥179.77**	¥0.67	**¥188.05**	+26%
Turkey (TRY)	¥6.87	¥90.57	¥29.19	¥0.74	¥7.07	**¥134.44**	¥0.73	**¥140.98**	-6%
China (CNY)	¥6.72	¥83.02	¥32.11	¥0.68	¥14.70	**¥137.23**	¥0.81	**¥149.77**	Base
India (INR)	¥6.87	¥78.50	¥20.43	¥0.64	¥5.46	**¥111.89**	¥0.84	**¥117.57**	-22%
Mexico (MXN)	¥7.25	¥90.57	¥33.86	¥0.74	¥7.35	**¥139.77**	¥0.76	**¥146.50**	-2%
United States (USD)	¥7.63	¥128.31	¥151.78	¥1.05	¥19.53	**¥308.30**	¥0.70	**¥321.46**	+115%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

2.07 kgCO2e · 4.12 /kg · cradle-to-gate

Material	**1.21 kgCO2e**	Material factor	**2.10 kg/kg (Steel)**
Process	**0.77 kgCO2e**	Grid intensity	**0.550 kg/kWh · 1.41 kWh**
Logistics	**0.08 kgCO2e**	Total	**2.07 kgCO2e**

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

§11 — Cost Intelligence Insights

3 findings · ~11% combined saving

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 55.4% of total is above the benchmark range of 18–35%. Cycle time or machine rate is elevated.
Impact	High · up to 10% potential saving
Benchmark	Process %: yours 55.4% vs industry 18–35%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixture time

[WARNING] Low material utilisation — high scrap rate

Finding	Material utilisation at 50% is below the machining benchmark of 72%. You are paying for 100% more raw material than ends up in the part.
Impact	Medium · up to 4% potential saving
Action 1	Redesign blank/preform geometry to minimise offcuts
Action 2	Optimise nesting/layout for sheet or profile stock

[INFO] Material cost is below benchmark range

Finding	Material at 4.5% is below the typical 30–55% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 4.5% vs industry 30–55%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

8 opportunities · combined ¥26.96/part (~18%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/part	%	Timeframe	Risk
1	Material	**Improve Material Utilisation via Near-Net-Shape**	Current utilisation at 50%. Near-net-shape process or improved nesting targets 80–90%.	**¥22.47**	15.0 %	Medium Term	Medium
2	Material	**Scrap Revenue Recovery at Index Prices**	50% of the bought material leaves as chips, offcuts or skeleton. That scrap has an index price — the credit belongs in the piece price, not in the supplier's margin.	**¥5.99**	4.0%	Quick Win	Low

#	Category	Action	Why it is on the list	Save/part	%	Timeframe	Risk
3	Process & Automation	**Consider near-net-shape pre-form (casting/forging)**	Process cost dominates (>40%) — Process cost at 55.4% of total. Machining conversion cost is unusually high.	**¥11.98**	8.0%	Medium Term	Medium
4	Process & Automation	**Unmanned / Lights-Out Running**	Labour is 21.4% of cost on a machine-paced part with healthy OEE (88%) — the classic profile for bar-feed or pallet-pool unattended running.	**¥8.99**	6.0%	Medium Term	Medium
5	Process & Automation	**Multi-Machine Manning**	Operations are manned at up to 1.0 operators per machine while the machine paces the cycle. One operator tending two machines is the textbook correction.	**¥7.49**	5.0%	Quick Win	Low
6	Design & Geometry	**Part-Count / Feature Integration Review (Boothroyd)**	5 operations suggest features or joined parts that may not need to exist separately. The Boothroyd minimum-part-count test asks of each: does it move, must it be a different material, must it be separate for assembly?	**¥8.99**	6.0%	Long Term	Medium
7	Design & Geometry	**Tolerance and Surface-Finish Relaxation**	Machining is 55.4% of the part. The tightest tolerance and finish callouts set the machine, the cycle and the inspection burden — reviewing them is the cheapest cost lever that exists.	**¥5.99**	4.0%	Quick Win	Low
8	Commercial & Sourcing	**Learning-Curve Price-Down Agreement**	Labour is 21.4% of cost and no learning curve is priced in. On a multi-year programme, labour productivity improves whether or not the contract captures it — the contract should.	**¥4.49**	3.0%	Quick Win	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Material	2	Improve Material Utilisation via Near-Net-Shape	**¥22.47**	¥28.46
Process & Automation	3	Consider near-net-shape pre-form (casting/forging)	**¥11.98**	¥28.46
Design & Geometry	2	Part-Count / Feature Integration Review (Boothroyd)	**¥8.99**	¥14.98
Commercial & Sourcing	1	Learning-Curve Price-Down Agreement	**¥4.49**	¥4.49

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Multi-Machine Manning
- Scrap Revenue Recovery at Index Prices
- Tolerance and Surface-Finish Relaxation
- Learning-Curve Price-Down Agreement

Long-Term Changes — Strategic Investment

- Part-Count / Feature Integration Review (Boothroyd)